

Unconditional Cash and Breastfeeding, Child Care, and Maternal Employment among Families with Young Children Residing in Poverty

LAURA STILWELL

Duke University

MARITZA MORALES-GRACIA

California State University, Los Angeles

KATHERINE MAGNUSON

University of Wisconsin–Madison

LISA A. GENNETIAN

Duke University

MARIA SAUVAL

University of California, Irvine

NATHAN A. FOX

University of Maryland

SARAH HALPERN-MEEKIN

University of Wisconsin–Madison

HIROKAZU YOSHIKAWA

New York University

KIMBERLY G. NOBLE

Columbia University

Electronically published May 22, 2024

Social Service Review, volume 98, number 2, June 2024.

© 2024 The University of Chicago. All rights reserved. Published by The University of Chicago Press. <https://doi.org/10.1086/729364>

ABSTRACT Poverty interferes with parents' breastfeeding, child-care, and employment options and ability to meet their parenting goals. This study—the first randomized controlled trial of early childhood poverty reduction in the United States—investigates how increased economic resources affect 1,000 low-income US mothers' breastfeeding, child-care, and employment practices and the ability to meet their intentions for these practices in the first year of their infant's life. The likelihood and length of breastfeeding, use of nonparental child care, and maternal employment did not statistically differ among mothers who received a high (\$333) or low (\$20) monthly unconditional cash gift. The higher monthly cash gift, however, delayed the starting age of child care by almost 1 month and increased mothers' ability to meet their breastfeeding intentions reported at birth.

Parents are faced with difficult choices when they try to balance breastfeeding, child care, and employment. In the United States, these challenges are especially hard for families with low incomes during children's infancy, when child-care costs are high and mothers often receive insufficient support for breastfeeding (Dinour and Bai 2016; Goldin, Kerr, and Olivetti 2022). In addition, parents who work in low-wage jobs often do not have paid parental or sick leave (Gennetian and Gassman-Pines 2022). Cash assistance may give low-income families flexible resources to help them care for their infant; however, most social welfare benefits in the United States are targeted, in-kind benefits rather than cash, such as subsidies for child care. The largest cash transfers to low-income families with children are provided through the tax system and are contingent on parental employment (Bitler, Hoynes, and Kuka 2017; Hoynes and Schanzenbach 2018). As a result, the impacts of unconditional cash support on low-income mothers' breastfeeding choices, use of nonparental child care, and paid employment are unclear. However, federal, state, and local governments and organizations are increasingly interested in how unconditional cash support may assist families and communities, as evidenced by the political efforts to increase the child tax credit and make it refundable, as well as the surge in the number of basic income and cash transfer pilot and demonstration programs underway. As of 2023, over 100 guaranteed income pilots were underway or concluded in the United States (Stanford Basic Income Lab 2023).

Understanding how unconditional cash assistance affects breastfeeding, child care, and maternal employment is important given the potential implications for both mothers' and children's health, as well as its potential to alleviate some of the more challenging constraints faced by low-income

families in caring for their children. Breastfeeding has been associated with a range of critical child and maternal health benefits such as reductions in infant mortality, asthma, and childhood obesity, as well as some types of cancer among mothers (Fitzsimons and Vera-Hernández 2022; Meek and Noble 2022). High-quality early child care has been linked to improved school readiness, particularly for children from more disadvantaged backgrounds (Magnuson and Shager 2010; Johnson 2017). On the other hand, mothers' returning to work shortly after birth has been associated with a range of negative outcomes, including increased child externalizing behaviors, lower school readiness, and increased maternal reports of depressive symptoms (e.g., Brooks-Gunn, Han, and Waldfogel 2002; Berger, Hill, and Waldfogel 2005; Herbst 2017). The ability of mothers to meet their parenting goals in these domains and have greater agency in constructing family routines may have positive impacts on their own well-being, as well as that of their family (e.g., Borra, Iacovou, and Sevilla 2015).

In this article, we use data from the first randomized controlled trial of early childhood poverty reduction in the United States—Baby's First Years—to assess the effect of increased economic resources on breastfeeding, nonparental care, and maternal employment. Specifically, we examine the impact of an unconditional, predictable monthly cash transfer provided to racially and ethnically diverse mothers residing in poverty at the time of their child's birth. We examine the cash gifts' effects on breastfeeding initiation and length, use of nonparental care, and maternal employment over their infant's first year of life. In addition, with data on mothers' intentions at the time of birth, we examine the alignment of mothers' breastfeeding and employment behaviors during their infant's first year with their stated intentions. We also assess whether impacts on meeting breastfeeding and maternal employment intentions differ by whether the infant was the mother's first birth, by the mother's employment status during pregnancy, and by the presence of other adults in the household.

THEORY AND LITERATURE REVIEW

Caring for an infant is costly and time intensive, and the ways in which parents rear their children is shaped by economic and sociocultural factors. Economic models of the family theorize that parents make decisions about their employment and infant care according to their preferences, subject to time and financial constraints (e.g., Blau and Currie 2006). Put simply, they

weigh the benefits and costs of parental care (time with the child, forgone wages) with those of parental employment and nonparental care (forgone time with the child, wages) and choose to work when the costs of staying home exceed the benefits. Economic theory would predict that by increasing the overall economic resources in the household, mothers who receive unconditional cash transfers might work less and thus use less nonparental care and possibly initiate breastfeeding more frequently or sustain it for longer.

For families with limited economic resources, economic theory predicts that family budgets constrain employment, breastfeeding, and child-care options. For parents with low incomes and small savings, the absence of paid leave policies often necessitates a return to paid work early in their infant's life. This timing is especially true for Black mothers, who have always had higher employment rates than mothers in other racial and ethnic groups and who are more likely to be single parents (Goldin 1977; Livingston 2018). Indeed, nearly 80 percent of Black mothers are the only employed adult in their families (Anderson 2016). Patterns of breastfeeding are strongly linked to patterns of maternal employment and family income: mothers with incomes below the federal poverty line (FPL) are less likely than higher-income mothers to initiate and continue breastfeeding (CDC 2023). This disparity is likely partially explained by maternal employment, which in turn limits low-income mothers' ability to breastfeed their children (Stilwell and Gennetian 2022). Mothers who return to work full-time and those who return shortly after giving birth are less likely to initiate breastfeeding and do so for a shorter time (e.g., Berger et al. 2005; Chatterji and Frick 2005; Mirkovic, Perrine, and Scanlon 2016). In addition, low-income women and women of color are overrepresented in industries that are often excluded from worker protections for breastfeeding (Yearby 2020; US Department of Labor 2023a).

Maternal employment outside the home often necessitates nonparental child care. However, with child care costing several thousand dollars a year, most low-income families cannot afford to pay market prices unless they receive government assistance, which is hard to access because of waiting lists and administrative burdens (Prenatal-to-3 Policy Impact Center 2022). Moreover, the prevalence of nonstandard or unpredictable work hours in the low-wage labor market makes finding child care that aligns with parents' work schedules difficult (Meyers and Jordan 2006; Henly and Lambert 2014; Enchautegui, Johnson, and Gelatt 2015). For these reasons, low-income families may opt to use care provided by friends and family

or home-based care, which often has more flexible hours (Chaudry 2004; Gennetian et al. 2019; National Survey of Early Care and Education Project Team 2021).

Economic theory fails to account for potentially important factors that affect parents' practices and decisions. Ecocultural frameworks argue that a wide range of individual, sociocultural, and contextual factors—such as beliefs, social network supports, and social norms—contribute to decisions related to employment, breastfeeding, and child care (Ajzen 1991; Meyers and Jordan 2006; Bronfenbrenner and Morris 2007). Meyers and Jordan (2006) argue that care arrangements reflect a process of accommodations to family needs, social and cultural expectations, available information, and other resources. For example, social networks serve as trusted resources that inform parents' feeding, child-care, and formal employment options (Hofferth et al. 1998; Gilbert 1998; Smith 2016; Lubbers, Small, and García 2020). Likewise, social norms affect the child-care and feeding practices that are perceived as acceptable substitutions for maternal care and breastfeeding (Weisner 2002; Lowe and Weisner 2004). Although the ecocultural framework underscores possible limitations of standard economic models of the family, because of its breadth, it fails to offer clear predictions for how unconditional cash might affect patterns of care for infants in a heterogeneous population. Nevertheless, we review some of the possible sociocultural and contextual factors that shape care practices among low-income communities of color in the United States.

Cultural norms shape breastfeeding patterns in the United States. From the late nineteenth century to mid-twentieth century, breastfeeding rates decreased dramatically because of a variety of social and economic factors, including the increasing need for women to work outside the home and the advent of pasteurized animal milk as an alternative to breast milk (Wolf 2003). This historical decline in breastfeeding means that many mothers today were not breastfed as infants, and thus they and their families may view formula feeding as the dominant social norm. In addition, because breastfeeding can be physically demanding, studies repeatedly show that partners' encouragement and support for breastfeeding is a strong predictor of initiation and continuation, which suggests another mechanism through which cultural beliefs about gender norms and breastfeeding may operate (Davidson and Ollerton 2020).

Because of historical and current structural racism and discrimination, Black and Hispanic families are disproportionately likely to have low

incomes. As a result, racial and ethnic disparities in breastfeeding rates are in part driven by economic and employment differences; however, they cannot be attributed to economic circumstances alone. In the United States, Hispanic infants have breastfeeding patterns similar to White, non-Hispanic infants (Gibson-Davis and Brooks-Gunn 2006). Although different patterns of infant feeding exist across Hispanic subgroups in the United States, generally, Hispanic cultural norms promote maternal breastfeeding, and Hispanic mothers with lower rates of acculturation have higher rates of breastfeeding initiation and duration (Gill 2009; Jones et al. 2015; Kim and Williams 2023).

Conversely, fewer Black, non-Hispanic infants are breastfed compared with other racial and ethnic groups, at least in part because of historical structural racism. Some qualitative research finds that Black mothers associate breastfeeding with the traumatic history of slaves being forced to wet-nurse slave owners' children, whereas other work points to the racialized, predatory behavior of formula manufacturers (DeVane-Johnson et al. 2018; Green, Killings, and Clare 2021). Finally, limited access to professional breastfeeding support also contributes to breastfeeding disparities; communities with higher percentages of Black populations are less likely to have health-care facilities that follow supportive breastfeeding practices (Lind et al. 2014).

Child-rearing beliefs, sociocultural norms, and community factors (e.g., the availability of affordable, high-quality child care) also influence choice of child-care arrangements. Parents are more likely to enroll their children in center-based programs if they believe learning how to share and take turns is highly important for children's school readiness (Liang, Fuller, and Singer 2000). Although it has been questioned (Crosby and Mendez 2016), some prior research has argued that familism, defined as strong values about the importance of family, drives Hispanic families to prefer their children to be cared for by a parent, if possible, and to use extended family and other informal forms of care (Stein et al. 2014; Shuey and Leventhal 2020). Therefore, as predicted by ecocultural framework, factors beyond economic resources matter for parents' child-care decisions. However, because economic factors are often confounded with contextual and cultural beliefs, it is hard to know which are more important contributors for which groups.

Providing a regular, reliable monthly cash support to low-income families may help families better balance both providing care for their children and meeting their material needs—cash can be used to pay for nonparental

child care or to offset forgone wages. The pandemic expansions of the child tax credit and the increase in basic income demonstration programs currently operating at the local level suggest a resurgence of interest in cash transfers as a policy tool (Constantino and Zeifman 2021; Stanford Basic Income Lab 2023). The extent to which cash support will change patterns of child care, employment, and breastfeeding is important to understand as part of these policy discussions.

In the US context, potential concerns about employment responses to unrestricted cash support have led to the creation of social safety net programs that include work requirements (Aizer, Hoynes, and Lleras-Muney 2022). In general, studies analyzing increases in unconditional income find null to small reductions in employment (see Marinescu 2018 for a review). More recently, multiple studies have examined the effect of the temporary expansion of the child tax credit between July 2021 and December 2021 and find no effects on parental employment (Pac and Berger 2024; Pilkauskas et al. 2022). However, no studies to date have specifically examined the impact of an income transfer on maternal employment in the first year after an infant's birth, when the benefits of staying home may be especially likely to outweigh forgone earnings. In addition, we would expect that any effects of an income transfer on work outside the home would also be associated with impacts on use of nonparental care, but this association has not explicitly been examined in prior research.

Research evaluating the effects of paid parental leave also informs our expectations about how a cash transfer to mothers of infants may affect maternal employment, breastfeeding, and child care. Generally, international studies find that generous paid leave policies lead to mothers being out of the labor market for a longer time after a birth, suggesting a longer time before they start using regular nonparental child care (e.g., Baker and Milligan 2008). If more generous leave policies reduce the time mothers are working in the first year of life, we would also expect that these policies might increase the duration of breastfeeding. In fact, several studies have demonstrated that increases in paid leave policies led to, or were associated with, increases in breastfeeding duration (e.g., Baker and Milligan 2008; Chai, Nandi, and Heymann 2018; Pac et al. 2019). Still, paid leave benefits are contingent on maintaining employment and therefore do not reflect the potential impacts of unconditional cash transfers.

In summary, parents face the complicated tasks of figuring how to structure their roles as caregivers and workers in the paid labor force (Gennetian

and Gassman-Pines 2022). For parents with low incomes, the challenge is made harder by the precarious nature of much low-wage work, the difficulties of combining breastfeeding and working outside of the home, and the high costs of child care. One approach to supporting parents of infants is providing additional financial resources that may substitute for forgone earnings or be used toward compensating others for nonparental child care. This study is among the first to examine whether regular, reliable cash transfers provided to mothers with low incomes during a child's infancy will affect their caregiving practices and employment.

This study also explores whether cash support affects mothers' ability to meet their breastfeeding and employment intentions and whether effects differ based on birth order, the mother having participated in the labor force while pregnant, or the mother and child living with another adult. A stronger connection to the labor market or having family support may affect issues of access and the choices available to low-income mothers (Riley and Glass 2002; Meyers and Jordan 2006; Kimmel and Connelly 2007; Laughlin 2011). For example, mothers who worked while pregnant may have more realistic expectations of how labor force demands may affect caretaking practices (e.g., breastfeeding and child care). Those with older children may be more informed about child-care choices but may also face higher child-care costs if care is needed for all their children. In addition, mothers living with at least one other adult may receive more in-household support as they seek to meet their parenting intentions (e.g., cash, help watching children, help with chores).

THIS STUDY

The Baby's First Years (BFY) study is the first randomized controlled trial of poverty reduction in early childhood in the United States. Between May 2018 and June 2019, 1,000 mothers were recruited after giving birth in 12 hospitals across four metropolitan areas: New York City, New Orleans, Minneapolis and St. Paul, and Omaha (Noble et al. 2021). Mothers were eligible to participate if they reported household income below the FPL; were at least 18 years old; spoke English or Spanish; had a single-child, full-term pregnancy; were taking the infant home with them; and had an infant without any known development or neurological problems who did not need intensive care. Once eligibility was determined, mothers were

invited to participate in a longitudinal study about the experiences of families with new babies. If they consented to participate, they received the baseline survey.

Following completion of the baseline survey, mothers were randomized to receive either a large (\$333) unrestricted monthly cash gift (treatment group; $n = 400$ mothers) or a nominal (\$20) unrestricted monthly cash gift (comparison group; $n = 600$ mothers). The high-cash gift represents an approximate 18 percent increase in the average annual baseline income of the age-1 sample. Mothers receive the monthly cash gifts via an automatically refilled debit card labeled “4MyBaby” until their child is just over 6 years old (76 months, extended from just over 4 years), pledged to mothers at the time of their child’s birth. The last payments are scheduled to be disbursed in 2024 and 2025. The money is refilled each month on the date corresponding to the child’s birthdate (e.g., the seventh day of every month for a child born on June 7), and the card can be used for direct purchases at vendors or to withdraw cash from ATMs. Investigators ensured that this money would not reduce the amount of public benefits that families are qualified to receive (e.g., Temporary Assistance for Needy Families, Supplemental Nutrition Assistance Program, child-care subsidies).

SAMPLE

The sample for this article comprises 931 mothers who completed the first in a series of follow-up surveys approximately 1 year after their child’s birth (low cash $n = 548$; high cash $n = 383$). The age-1 sample is diverse along several characteristics, including race and ethnicity, number of previous children, marital status, and household structure (as shown in table A1 [available online]). About 41 percent of mothers self-identified as Black and non-Hispanic; 42 percent reported that they were Hispanic. For fewer than a third (28 percent), the infant was their firstborn. Slightly less than half (45 percent) of the mothers were never married; one out of five reported being married. Approximately 39 percent of mothers reported the biological father living in the same household, while in slightly over half of households (54 percent), the mother was the only adult resident. The average household income was \$21,769, which was slightly above the FPL for a family of three in 2019 (\$21,330). More than half of mothers (57 percent) worked while pregnant, and 15 percent continued working up to their infant’s birth.

DATA AND MEASURES

The age-1 survey collected information about the participants' socioeconomic circumstances, household environment, and family life, including breastfeeding patterns, use of nonparental child care, and maternal employment history. Because the study recruitment occurred over a 13-month period, age-1 interviews were completed between July 2019 and June 2020. The COVID-19 pandemic began during age-1 data collection. Though data collection was originally administered in person during a home visit prior to the pandemic, we shifted to phone interviews on March 14, 2020. At that time, 65 percent of the high-cash gift group and 57 percent of the low-cash gift group had already completed the age-1 survey in person.

The baseline survey collected information about mothers' intentions at the time of the child's birth with respect to breastfeeding and employment.¹ The age-1 survey included retrospective information about their experiences with employment and breastfeeding and their child's experiences of home-based or center-based care. The age-1 survey also asked mothers about their employment within the past month and child-care arrangements within the past week. Table 1 shows that most mothers intended to breastfeed and work after giving birth, as measured during the baseline survey. In addition, most mothers reported breastfeeding, and slightly more than half worked during their child's first year. Figure 1 demonstrates patterns of breastfeeding, child care, and maternal employment by infant age, with breastfeeding declining each month and use of home-based or center-based child care and maternal employment steadily increasing each month. We provide more details about each of these measures below.

Intentions

The baseline survey asked mothers whether they intended to breastfeed and for how long. The interviewers also asked mothers whether they planned to return to work (if they had worked during pregnancy) or to begin

1. Mothers were also asked about their child-care intentions and could indicate multiple child-care arrangements (e.g., partner or father of child, relatives, nonrelatives, center care); however, the question did not specify child care in the first year and did not ask when they planned to start each child-care arrangement. Consequently, we cannot know if an intention to have a specific type of child care (e.g., parental vs. nonparental) corresponds to the first year of life, and, therefore, we do not include these variables in our analysis.

TABLE 1. Descriptive Statistics of Intentions at Birth and Practices in the First Year: Age-1 Baby's First Years Sample

	Full Sample Mean (SD)	Low- Cash Gift Mean	High- Cash Gift Mean	Standardized Mean Difference	<i>p</i> Value	<i>N</i>
Baseline intentions:						
Intend to breastfeed (%)	.77	.78	.75	-.10	.38	928
Anticipated breastfeeding length (mos.)	9.58 (8.05)	9.24 (6.68)	10.10 (9.72)	.11	.20	682
Intend to work for pay (%)	.82	.82	.83	.04	.49	846
Months after birth mom plans to start working	2.89 (2.97)	2.98 (3.29)	2.77 (2.46)	-.07	.37	711
Intend to breastfeed but not work (%)	.15	.16	.14	-.10	.65	843
Intend to work but not breastfeed (%)	.21	.20	.22	.07	.31	843
Intend to breastfeed and work (%)	.62	.62	.61	-.03	.80	843
Intend to not breastfeed or work (%)	.03	.03	.02	-.25	.50	843
Age-1 practices:						
Ever breastfed (%)	.77	.77	.76	-.03	.60	930
Infant age when stopped breastfeeding (mos., conditional on breastfeeding)	3.04 (2.54)	2.94 (2.43)	3.20 (2.68)	.10	.23	579
Ever used home-based or center-based care (%)	.41	.39	.44	.12	.16	930
Infant age when started home-based or center-based care (mos., conditional on using this type of care)	4.58 (3.11)	4.23 (3.05)	5.03 (3.14)	.26	.00	383
Ever worked for pay (%)	.54	.54	.54	.00	.91	928
Number of months mom worked (conditional on working)	6.71 (3.57)	6.75 (3.66)	6.66 (3.44)	-.03	.68	504

Note.—The *p* values were derived from a series of ordinary least squares bivariate regressions in which each respective baseline intention was regressed on the treatment status indicator using robust standard errors and site-level fixed effects. Standardized mean differences were calculated using Hedges *g* for continuous variables and Cox index for dichotomous variables. We conducted a joint test of orthogonality using a probit model with robust standard errors and site-level fixed effects to test whether all baseline intentions are different between high- and low-cash gift groups. Joint test: chi-square (8) = 8.77, *p*-value = .27, *n* = 931.

working (if they had not worked during pregnancy). If they replied yes to either question, the survey asked when they intended to start working. Of the respondents, 79 reported they planned to return to work or begin working but did not specify when. We coded these values as missing for the variable “intended to work,” since we do not know whether they intended to return to work or begin working within the child’s first year.

We find that just over three-fourths of mothers (77 percent) had planned to breastfeed, and the majority (82 percent) had planned to work for pay in the first year. Most mothers who planned to work also intended to breastfeed (62 percent). Among mothers who intended to breastfeed,

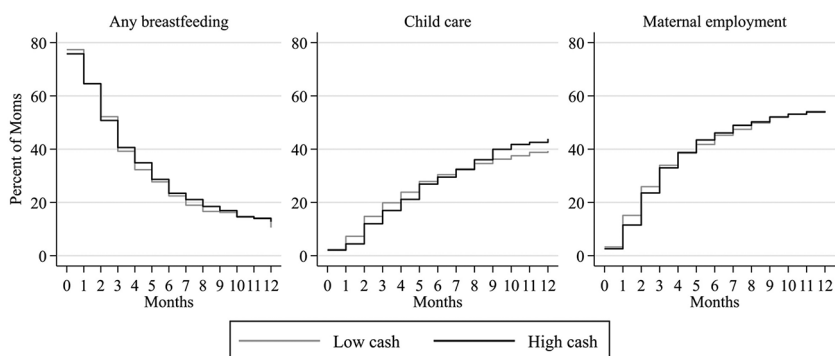


FIGURE 1. Breastfeeding, use of home-based or center-based child care, and maternal employment by infant age. “Child care” refers to use of home-based or center-based care only. For the child care graph, the child age (x axis) represents the first time the child used home-based or center-based child care. For maternal employment, the child age represents the time at which the mother began working during the first year. It is possible that the family stopped using home-based or center-based child care or that the mother stopped working in the months that followed.

they reported intending to do so for an average of 9.6 months. Among mothers who planned to work for pay in the first year, on average, they planned to do so when the baby was close to 3 months of age. Mothers’ baseline intentions did not differ by their cash-gift group assignment. (See table 1 for full descriptive statistics on intentions.)

Age-1 Breastfeeding Practices

The survey asked mothers if they ever breastfed and, if so, in which months. As a result of the timing of the age-1 survey, 5 mothers reported that they stopped breastfeeding when the child was older than 12 months, and 127 mothers reported still breastfeeding at the time of the age-1 survey. We consider these values to be missing for the variable that indicates the number of months mothers breastfed. The estimates provided below remain largely unchanged if these values are included and truncated at age 12 months. There were no differences in the proportion of high- or low-cash gift mothers who were still breastfeeding at age 1.

About 77 percent of mothers in our sample reported having breastfed their baby, which is close to the national average for mothers with incomes below the FPL (CDC 2023). The average length of breastfeeding was 3.04 months. (See table 1 for full descriptive statistics on practices.)

Age-1 Nonparental Child Care

The survey asked mothers if their child had ever spent 5 or more hours with a nonrelative who cared for the child in the nonrelative's home (home-based care) and if the child had spent any time in center-based care (i.e., a day care or child-care center). For each of these questions, when the mother responded in the affirmative, she was asked in which month the child began this type of care. Fewer than half of mothers used home-based or center-based child care in the first year of their child's life (41 percent), which is in line with national averages (Swenson and Simms 2020).

The survey also asked mothers whether, during the last week, the child (1) spent 5 or more hours in home-based care, (2) spent 5 or more hours in child care or day care, and (3) spent any time in nonparental care, including with caregivers such as babysitters, grandparents, siblings, or other related or unrelated adults. Mothers who responded in the affirmative to any of these questions were asked how many total hours the child spent in nonparental care that week.

Age-1 Maternal Employment

The survey asked mothers whether they had worked for pay or in self-employment in the first year of the child's life. If they had, the interviewer also asked in which months they worked and the infant's age when the mother started working during the first year. Slightly over half (54 percent) of our sample ever worked for pay in the first year following their child's birth, which is consistent with national averages for mothers with our sample's average education level (Knop 2019). These mothers worked, on average, for 6.7 months. In addition, the survey asked whether they were currently working for pay and, if so, how many hours they worked per week in the month before the age-1 interview.

Age-1 Met Intentions

Using the baseline and age-1 measures, we created variables for whether mothers had met their stated intentions at baseline. For example, we considered mothers to have met their intention to breastfeed if they stated at baseline that they planned to breastfeed and reported ever breastfeeding by age 1. Similarly, mothers were considered to have met their intention not to breastfeed if they reported not planning to breastfeed at baseline and reported never breastfeeding in the year following their infant's birth. The variables for meeting the intention to work or not to work were constructed

similarly, using mothers' baseline reports of plans to work in the first year (either return to work or start working) and age-1 measures of working for pay or self-employment since the infant's birth.

METHODS

The following regression model was used to estimate the impact of the high-cash gift on breastfeeding practices, use of nonparental child care, maternal employment patterns, and meeting intentions for breastfeeding and employment:

$$Y_i = \pi \text{HighCashGift}_i + \mathbf{X}_i\beta + \mathbf{S}_{\text{Site}} + \varepsilon_i. \quad (1)$$

The variable Y_i is the outcome of interest for mother i , and HighCashGift_i is an indicator of whether a mother is in the high-cash gift group; π is the estimate of being assigned to the high-cash gift group.

Because individual characteristics such as maternal age, education attainment, and number of prior children can affect parents' care and work decisions (Riley and Glass 2002; Meyers and Jordan 2006), we included a vector of baseline covariates, $\mathbf{X}_i$, consisting of maternal characteristics (mother's age, highest level of education attained, race and ethnicity, marital status, general health, an indicator of maternal depressive symptoms, cigarette and alcohol consumption during pregnancy) and household characteristics (number of other children born to the mother, number of adults in the household, infant's biological father living with the mother, household income, household net worth). The vector also included infant's birth characteristics (sex, birth weight, gestational age at birth), an indicator for the switch from in-person to phone interviews resulting from the onset of COVID-19, and child's age at interview (in months). Given that access to lactation support may increase mothers' likelihood of breastfeeding, the vector of baseline covariates $\mathbf{X}_i$ also includes a variable for whether the infant's birth occurred in a Baby-Friendly Hospital, a designation that hospitals can receive for adhering to a set of practices that help mothers initiate and continue breastfeeding (Baby-Friendly USA 2023). The vector $\mathbf{S}_{\text{Site}}$ consists of dummy variables for each of the four sites because random assignment occurred within site. We used Huber-White robust standard errors to handle any heteroskedasticity.

For each of our outcome measures, we calculated the minimum detectable effect at the 0.05 alpha level on a two-tailed test with 80 percent power (see table 2). Given the successful implementation of the debit card and nearly universal take-up (Halpern-Meekin et al. 2024; Gennetian et al. 2023), this

TABLE 2. Estimated Minimum Detectable Impact of Baby’s First Years High-Cash Gift for the Age-1 Sample

	Low-Cash Comparison Group Mean	Minimum Difference Powered to Detect
Breastfeeding:		
Ever breastfed	77	7.4 percentage points
Infant age in months when stopped breastfeeding	2.94 months	19.2 days
Breastfed for 6 months or longer	31	10.1 percentage points
Child care:		
Ever use home-based or center-based care in first year	39	9.4 percentage points
Infant age in months when started home- based or center-based care	4.29 months	28.5 days
Used nonparental child care in the last week	46	9.5 percentage points
Hours spent in nonparental child care last week	7.96 hours	2.6 hours
Maternal employment:		
Ever worked for pay since birth	54	9.4 percentage points
Number of months worked	6.75 months	28.7 days
Working for pay at time of age-1 survey	45	9.3 percentage points
Number of hours worked for pay per week in past month	14.65 hours	3.5 hours

estimate can be interpreted as the treatment-on-the-treated estimate and captures the impact of a net positive income increase of \$313 per month.

To analyze whether effects on breastfeeding and employment differ based on the mothers’ baseline intentions to breastfeed and to work, respectively, we estimated the following specification:

$$Y_i = \pi \text{HighCashGift}_i + \theta(\text{HighCashGift}_i \times \text{Intention}_i) + \gamma \text{Intention}_i + \mathbf{X}_i \beta + \mathbf{S}_{\text{Site}} + \varepsilon_i, \tag{2}$$

where Y_i , HighCashGift_i , $\mathbf{X}_i$, and $\mathbf{S}_{\text{Site}}$ are the same as those for equation (1), and robust standard errors were included. Intention_i is an indicator for whether the mother intended to perform the behavior that corresponds to the outcome. Therefore, π represents the causal estimate of receiving the high-cash gift group among those who did not intend to perform the behavior, and $\pi + \theta$ corresponds to the impact of the high-cash gift among those who did.

We explored effects of the cash-gift intervention on meeting breastfeeding and employment intentions (regardless of what they were) by three baseline characteristics of interest: whether the focal child was the first child born to the mother, whether the mother worked while pregnant,

and whether other adults resided in the household. We used the following regression model:

$$Y_i = \pi \text{HighCashGift}_i + \theta(\text{HighCashGift}_i \times Z_i) + \gamma Z_i + \mathbf{X}_i\beta + \mathbf{S}_{\text{Site}} + \varepsilon_i, \quad (3)$$

where the notation is the same as equation (2) with the exception that Z_i corresponds to an indicator for first child, maternal employment while pregnant, or no other adults in the household.

Although differential attrition is an issue in many randomized controlled trials, a high overall response and good balance across the high- and low-cash gift groups in our age-1 sample suggest that differential attrition is not an important concern for these analyses. We tested and confirmed baseline equivalence for our age-1 sample by regressing treatment status on a host of measures collected prior to treatment assignment, including infant characteristics, mother's age and demographics, mother's age and ethnicity, mother's marital status, mother's health, household characteristics, and mother's employment prior to her infant's birth (table A1; for table showing equivalence of full baseline sample, see Noble et al. 2021). Nevertheless, to address the possibility that participants who completed the age-1 survey ($n = 931$) could be different from the 69 participants who did not respond to the age-1 survey in a way that is correlated with the treatment, we used inverse probability treatment weights to adjust the age-1 sample to reflect the characteristics of the full study sample at baseline and to address potential attrition-driven imbalance between groups (see table A2).

COVID-19 CONTEXT

Our results could be affected by the COVID-19 pandemic, which began during age-1 data collection. First, the context of the pandemic may have influenced mothers' willingness or ability to respond to the survey. After the onset of the pandemic, the survey was administered by phone, which may have made participation easier. This ease, in turn, may have resulted in differences in the characteristics of the mothers who responded before versus after this procedural change. A comparison of baseline characteristics of the prepandemic and pandemic-era subsample show that both groups are statistically similar in all dimensions, except that there were more married mothers among those who completed the age-1 survey after the switch to phone surveys on March 14, 2020 (Gennetian et al. 2022).

A second issue is whether the pandemic's impacts on daily life, including closure of child-care facilities and temporary layoffs, might have altered the effect of the cash gift.² Fortunately, most of the survey questions for breastfeeding, child care, and employment were retrospective, asking about the first 12 months of the babies' lives, which corresponded to a pre-COVID-19 scenario. This is because, though the age-1 survey had been planned around the child's first birthday, some mothers responded to the survey later (the average age of the infant at time of the survey was 13 months old, with a standard deviation of 2 months). For instance, if a participant responded to the survey in the beginning of May 2020 and the baby was 14 months old, the number of months she reported having worked during her child's first year was not affected because the focal child turned 1 before the pandemic started. In fact, although 35 percent of the age-1 sample mothers were interviewed after March 14, 2020, a large proportion of them had babies older than 12 months. In contrast, outcomes that were measured at the time of the interview (e.g., whether the mother is currently working for pay, whether they used nonparental care in the last week) reflect the combination of prepandemic and pandemic-era conditions.

RESULTS

Did the provision of reliable, regular monthly cash transfers affect patterns of breastfeeding, child care, and employment for mothers and their newborns? Comparing the high-cash gift group with the low-cash gift group, we found no differences in breastfeeding rates or length, use of home-based or center-based care, hours spent in nonparental care in the last week, or measures of maternal employment (table 3). However, among those who used home-based or center-based care, the low-cash gift group was more likely to enter formal child care at a younger age. Specifically, low-cash gift group infants entered home-based or center-based care 1 month sooner than the high-cash gift group infants ($p < .01$). The results in table 3 are robust to the exclusion of covariates and the use of inverse probability analytic weights that account for attrition (table A2).

2. The sample included 218 mothers who had babies under 12 months old when the pandemic started. Most of them only experienced the COVID-19 pandemic in the last month before the interview ($n = 131$); 71 mothers were affected in months 11 and 12; and 16 mothers were affected in months 10, 11, and 12.

TABLE 3. Summary of Estimates of the Impact of Baby's First Years High-Cash Gift on Breastfeeding, Child Care, and Employment in the First Year

	Low-Cash Gift Mean (1)	OLS ITT with FE and Covariates (2)	Effect Size (3)	<i>p</i> Value (4)	Adjusted <i>p</i> Value (5)	<i>N</i> (6)
Breastfeeding:						
Ever breastfed (%)	.77	.00 (.03)	-.01	.88	1.00	930
Infant age in months when stopped breastfeeding	2.94	.21 (.23)	.08	.36	.96	579
Breastfed for 6 months or longer (%)	.31	-.02 (.04)	-.04	.64	1.00	706
Child care:						
Ever use home-based or center-based care in first year (%)	.39	.04 (.03)	.09	.19	.84	930
Infant age in months when started home-based or center-based care (conditional on using this type of care)	4.23	.97** (.33)	.32	.00	.05	383
Used nonparental child care in the last week (%)	.46	.00 (.03)	.00	1.00	1.00	930
Hours spent in nonparental child care last week	7.96	-.07 (.93)	-.01	.94	1.00	926
Maternal employment:						
Ever worked for pay since birth (%)	.54	.00 (.03)	.00	.95	1.00	928
Number of months worked (conditional on working)	6.75	.10 (.34)	.03	.76	1.00	504
Working for pay at time of age-1 survey (%)	.45	-.04 (.03)	-.09	.18	.84	931
Number of hours worked for pay per week in past month	14.65	-1.95 (1.24)	-.10	.11	.71	922

Note.—FE = fixed effects; ITT = intention to treat; OLS = ordinary least squares. Standard errors are in parentheses. Effect sizes (col. 3) correspond to site-level fixed effects and covariates models. Covariates from baseline survey: Mother's Age, Completed Schooling, Household Income, Net Worth, General Health, Mental Health, Race and Ethnicity, Marital Status, Number of Adults in the Household, Number of Other Children Born to the Mother, Smoked during Pregnancy, Drank Alcohol during Pregnancy, Father Living with the Mother, Child's Sex, Birth Weight, Gestational Age at Birth, and Birth Took Place in a Baby-Friendly Hospital. To prevent losing observations, baseline covariates that were missing were replaced with the covariate's mean, and we included a dummy to indicate whether the covariate is missing. Other covariates: Phone Interview, Child Age at Interview (in months). Adjusted *p* value refers to adjusting for multiple hypothesis testing with Westfall and Young's (1993) step-down resampling methods. Because of the timing of the age-1 survey, five mothers reported that they stopped breastfeeding when the child was older than 12 months. These five entries are treated as missing for infant age in months when stopped breastfeeding.

** *p* < .01.

Absent differences in employment, it was surprising to see the difference in the age of entrance into formal child care. We therefore also explored whether the younger age of entry into formal child care for the low-cash gift group could be accounted for by hours of employment (part

time vs. full-time). Among mothers who worked in the first year, we found no differences between the high- and low-cash groups regarding the age of infants when mothers started employment, either part time or full-time. In addition, the high-cash gift group was not more likely to be currently working full-time or currently working part time in the month prior to the age-1 survey (results available on request).

The study was conducted in four sites that differed in the range and generosity of policies to support mothers with young children. Specifically, New York City has more generous state policies related to paid leave and support for breastfeeding in the workplace.³ For this reason, we compared the effects of the high-cash gift in New York City with the other three sites (New Orleans, Minneapolis and St. Paul, and Omaha). The high-cash gift in New York was associated with a marginally statistically significant increase in ever breastfeeding (5 percentage points, $p < .10$), decrease in working for pay at the time of the age-1 survey (-12 percentage points, $p < .05$), and decrease in the number of hours worked in the past week (-5.2 hours, $p < .01$). For each of the outcomes, the effect of the high-cash gift in New York differed statistically from the treatment effect in the other sites ($p < .10$). These findings suggest that the high-cash gift might have had a larger impact in a city with greater policy supports for new mothers. However, the New York sample and context differed in several other ways—it had a higher percent of Hispanic and married mothers, a higher cost of living, and more generous public benefits. Therefore, these differences cannot be directly attributed to the city's leave and breastfeeding policies.

One reason we may not have found effects of the high-cash gift on breastfeeding and employment outcomes in the full sample is that mothers had heterogeneous intentions at baseline. To explore this, we tested whether the high-cash gift group was more likely to have met their stated intentions regarding breastfeeding and employment (table 4). Regardless of cash-gift group assignment, baseline intentions strongly predicted behaviors. Mothers who intended to breastfeed were more likely to have ever breastfed and to have breastfed for longer, whereas mothers who reported intending to

3. Since 2018, all qualified employees in New York State were eligible for paid leave with few exceptions (NYS 2023). Overall, New York's protections for breastfeeding at work were stronger than those at other BFY sites during BFY infants' first year (NCSL 2021), prohibiting employers from discriminating against employees who breastfeed or pump at work and requiring that employers allow for mothers to pump milk at work (New York State Department of Health 2017; US Department of Labor 2023b).

TABLE 4. Summary of Estimates of the Impact of the Baby's First Years High-Cash Gift on Breastfeeding, Child Care, and Employment in the First Year by Baseline Intentions

Breastfeeding Outcomes	Ever Breastfed		Infant Age When Stopped Breastfeeding (in Months)		Breastfed for 6 Months or Longer	
High-cash gift	.00 (.03)	-.08 (.05)	.21 (.23)	.26 (.48)	-.02 (.04)	-.03 (.07)
High-cash gift × Intended to breastfeed		.10 ⁺ (.06)		-.11 (.52)		.01 (.07)
Intended to breastfeed		.66** (.04)		1.94** (.42)		.27** (.06)
N	930	930	579	579	706	706

Employment Outcomes	Mom Ever Worked for Pay Since Birth		Number of Months Worked in the First Year		Number of Hours Worked for Pay per Week in Past Month	
High-cash gift	.00 (.03)	.00 (.07)	.10 (.34)	-.62 (1.14)	-1.95 (1.24)	-3.96 (2.47)
High-cash gift × Intended to work		.00 (.08)		.78 (1.21)		2.38 (2.88)
Intended to work		.37** (.05)		1.60 ⁺ (.82)		5.07* (2.00)
N	928	928	504	504	922	922

Note.—Standard errors are in parentheses. Ordinary least squares intention-to-treat models were run with site-level fixed effects and robust standard errors. Covariates from baseline survey included Mother's Age, Completed Schooling, Household Income, Net Worth, General Health, Mental Health, Race and Ethnicity, Marital Status, Number of Adults in the Household, Number of Other Children Born to the Mother, Smoked during Pregnancy, Drank Alcohol during Pregnancy, Father Living with the Mother, Child's Sex, Birth Weight, Gestational Age at Birth, and Birth Took Place in a Baby-Friendly Hospital. To prevent losing observations, baseline covariates that were missing were replaced with the covariate's mean, and we included a dummy to indicate whether the covariate is missing. Other covariates: Phone Interview and Child Age at Interview (in months). Outcome variable for hours mom worked in past week is top coded at the 99.5th percentile.

⁺ $p < .10$.

* $p < .05$.

** $p < .01$.

work in the first year were more likely to have ever worked for pay since birth, worked more months, and worked more hours for pay per week in the month prior to the age-1 survey. However, the high-cash gift did not increase the likelihood of working for pay among those who intended to work. The high-cash gift was marginally associated with increased breastfeeding among those who intended to breastfeed. In addition, the high-cash gift increased the likelihood that mothers met their breastfeeding intention regardless of what those intentions were (see table 5).

Given that formula is a substitute for breastmilk, we explored whether the cash gift was associated with the use of formula. Among mothers who

TABLE 5. Summary of Estimates of the Impact of the Baby's First Years High-Cash Gift on Ability to Meet Breastfeeding and Maternal Employment Intentions

	Low-Cash Gift Mean (1)	OLS ITT with FE and Covariates (2)	N (3)
Breastfeeding intentions:			
Intended to breastfeed and did	.93	.03 ⁺ (.02)	710
Intended to not breastfeed and did not	.75	.11 ⁺ (.06)	217
Met intention to breastfeed or not breastfeed	.89	.04* (.02)	927
Employment intentions:			
Intended to work and did	.64	-.02 (.04)	766
Intended to not work and did not	.76	.06 (.08)	150
Met intention to work or not work	.67	-.02 (.03)	844

Note.—FE = fixed effects; ITT = intention to treat; OLS = ordinary least squares. Standard errors are in parentheses. Ordinary least squares intention-to-treat models were run with site-level fixed effects and robust standard errors. Covariates from baseline survey included Mother's Age, Completed Schooling, Household Income, Net Worth, General Health, Mental Health, Race and Ethnicity, Marital Status, Number of Adults in the Household, Number of Other Children Born to the Mother, Smoked during Pregnancy, Drank Alcohol during Pregnancy, Father Living with the Mother, Child's Sex, Birth Weight, Gestational Age at Birth, Birth Took Place in a Baby-Friendly Hospital, Breastfeeding Intentions, and Employment Intentions. To prevent losing observations, baseline covariates that were missing were replaced with the covariate's mean, and we included a dummy to indicate whether the covariate is missing. Other covariates: Phone Interview and Child Age at Interview (in months).

⁺ $p < .10$.

* $p < .05$.

did not intend to breastfeed, the high-cash gift increased reports of ever using formula or other milks regularly in the first year by 7 percentage points ($p < .01$; results available on request). Likewise, because the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) is a common way for low-income mothers to obtain formula for their infants, we considered whether the high-cash gift mothers may have been more likely have access to formula through this program. We found no difference in WIC receipt between the high- and low-cash gift groups at age 1, including among the subsample who did not intend to breastfeed.

With respect to maternal work outcomes, mothers in the high- and low-cash gift groups were equally likely to have met their employment intentions. Only 64 percent of mothers who intended to work for pay in the first year (and 54 percent of all mothers) worked. Given this disparity between expressed intentions and actual employment, we explored whether baseline factors were correlated with mothers' ability to meet their work

intentions. Among those who intended to work for pay, mothers who continued working until birth or worked while pregnant were more likely to have successfully met this intention (estimates available on request). Other characteristics of mothers, such as the number of previous children, mother's age, marital status, and educational attainment, did not predict whether mothers met their intention to work.

We performed sensitivity analyses to test the robustness of our analysis of mothers' breastfeeding and employment intentions. Specifically, we estimated models without covariates and used inverse probability treatment weights to account for attrition and any minor covariate imbalance between the high- and low-cash groups (table A2). We found that the results were similar across all these alternative specifications.

In addition, we explored whether the impact of the high-cash gift on mothers' ability to meet their employment and breastfeeding intentions differed across specific groups of mothers who theoretically may have been more (or less) able to meet these intentions. We examined three infant and maternal characteristics: (1) birth order, specifically, whether the infant in this study is the firstborn; (2) employment status during pregnancy; and (3) the presence of another adult in the home. Table 6 shows that the effect of the high-cash gift did not differ across these groups.

DISCUSSION

The lack of paid parental leave and high cost of child care in the United States means that many low-income mothers face difficult decisions with complicated tradeoffs regarding breastfeeding, child care, and employment. Low incomes and lack of policy supports, especially economic assistance, constrain parents' options and may ultimately affect children's health and development, as well as the health and well-being of mothers. Therefore, it is important to understand whether providing low-income parents with additional income may result in changes in their care and employment decisions, especially if these changes better align with their family needs.

In this article, we examined whether a monthly, unconditional cash transfer affects mothers' breastfeeding, child-care, and employment decisions over their infant's first year of life and whether the cash transfer allows mothers to meet their stated intentions at the time of their infant's birth. Over three-quarters of the mothers participating in the BFY study intended to breastfeed, and a similarly high percentage of mothers indicated they

TABLE 6. Summary of Estimates of the Impact of the Baby's First Years High-Cash Gift on Ability to Meet Breastfeeding and Employment Intentions by Birth Parity, Maternal Connection to Labor Market, and Family Social Supports

	Met Intention to Work or Not Work (N = 844)	Met Intention to Breastfeed or Not Breastfeed (N = 927)
First child:		
High-cash gift	.00 (.04)	.04 (.02)
First child × High-cash gift	-.05 (.07)	.02 (.04)
First child	-.03 (.06)	-.02 (.04)
Mom worked while pregnant:		
High-cash gift	-.01 (.05)	.07* (.03)
Mom worked while pregnant × High-cash gift	-.02 (.07)	-.06 (.04)
Mom worked while pregnant	.19** (.05)	.04 (.03)
No other adults in household:		
High-cash gift	-.04 (.04)	.05* (.02)
No other adults in household × High-cash gift	.08 (.07)	-.04 (.04)
No other adults in household	.06 (.06)	.08+ (.04)

Note.—Standard errors are in parentheses. Ordinary least squares intention-to-treat models were run with site-level fixed effects and robust standard errors. Covariates from baseline survey included in all models: Mother's Age, Completed Schooling, Household Income, Net Worth, General Health, Mental Health, Race and Ethnicity, Marital Status, Number of Adults in the Household, Number of Other Children Born to the Mother, Smoked during Pregnancy, Drank Alcohol during Pregnancy, Father Living with the Mother, Child's Sex, Birth Weight, Gestational Age at Birth, Birth Took Place in a Baby-Friendly Hospital, Breastfeeding Intentions, and Employment Intentions. To prevent losing observations, baseline covariates that were missing were replaced with the covariate's mean, and we included a dummy to signify whether the covariate is missing. Other covariates: Phone Interview and Child Age at Interview (in months).

+ $p < .10$.

* $p < .05$.

** $p < .01$.

intended to work for pay during their child's first year of life. Approximately a year later, their actual maternal breastfeeding practices and employment rates fell far short of their intentions. Most important, the \$333 BFY cash gift given every month for the child's first year did not substantively change

mothers' patterns of working, breastfeeding, or use of nonparental child care—with two exceptions. First, the high-cash gift led to a small, but statistically significant, increase in the likelihood of mothers meeting their breastfeeding intentions (i.e., either to breastfeed or not). Second, among those who used home-based or center-based child care in the first year, mothers in the high-cash gift group delayed their use of these forms of child care by 1 month (i.e., from the fourth to the fifth month of the baby's life).

By considering the timing of mothers' return to work, this article adds to our understanding of how the BFY high-cash gift affected maternal employment. Prior work by Sauval and colleagues (2022) finds the BFY cash gift did not substantively affect maternal employment at child-ages 1–3 or maternal earnings.⁴ Taken together, these findings suggest that current political concerns that providing unconditioned cash or other public benefits of a similar amount to low-income families would substantially discourage work of new parents are unfounded (Katz 2013). Indeed, evidence from the recent expansion of the refundable child tax credit also shows that maternal employment was not reduced by these payments (Pac and Berger 2024; Pilkauskas et al. 2022).

The fact that we do not see an increase in formal child-care use among the high-cash gift group is not surprising. On average, formal child care in the United States costs approximately \$850 per month—far below the amount of the high-cash gift (Child Care Aware of America 2022). Mothers would likely only be able to afford market-based child care if they also received additional financial assistance. Nationally, only about 25 percent of eligible families receive a child-care subsidy (Prenatal-to-3 Policy Impact Center 2022); a similarly small fraction (22 percent) of BFY mothers report access to free or reduced child care at age 1.⁵

We were surprised that mothers receiving the high-cash gift delayed their child's entrance into formal home-based or center-based child care by a month compared with the low-cash gift group, given that there was no similar delay in entering the labor market or differences in patterns of part-time or full-time employment. One possible explanation is that mothers

4. Sauval and colleagues (2022) found a 27 percent, statistically significant reduction in the probability of working full-time and a 6-hour reduction in the hours worked per week at the time of the age-2 survey. These results were not detected at ages 1 or 3. The working paper explores the hypothesis that the age-2 findings may be a response to the COVID-19 pandemic.

5. No differences exist between the high- and low-cash gift groups.

relied on informal nonparental arrangements for younger infants in this group. For instance, faced with the need to return to work, mothers in the high-cash gift group might have been able to pay an older sibling, family member, or trusted friend to care for their children, which allowed them to delay the use of a home- or center-based child-care facility. However, we did not ask about the regular use of informal nonparental child care throughout the first year and therefore are unable to test this hypothesis.

Why might a regular, reliable cash gift affect mothers' ability to meet their intention of not breastfeeding? We find that among mothers who did not intend to breastfeed, the high-cash gift increased the likelihood their infants received any formula or other milks regularly in the first year. This pattern might explain why the cash gift is associated with mothers' meeting their breastfeeding intentions, even when their intention was to not breastfeed. Our interpretation aligns with previous research demonstrating that mothers perceive formula to be expensive (Guttman and Zimmerman 2000; Johnson 2019). Although mothers may have been able to get formula through WIC, we find no difference in WIC receipt between the cash-gift groups at age 1, including among those who did not intend to breastfeed.

This study also explored whether the effects of the high-cash gift on the ability of mothers to meet their breastfeeding and employment intentions would differ across subgroups defined by factors that influence care and employment practices in the first year—namely, whether the child is a first birth, maternal employment while pregnant, and the presence of another adult in the household. We found the ability to meet intentions was not more or less affected by the high-cash gift across these subgroups. Of course, the impact of the cash transfers may differ by many other factors we did not examine (e.g., across differing levels of father involvement).

In exploratory analyses, we found statistically significant effects of the high-cash gift in one site—New York City—on breastfeeding, working for pay at the time of the age-1 survey, and the number of hours worked in the past week. One possible explanation is that the differing policies on family leave and breastfeeding in the workplace made it easier for mothers to delay entry into the labor market, allowing them to breastfeed their children. However, because of differences across sites in the availability of social programs and services, costs of living, and sample demographic characteristics, it is unclear what is driving these differences.

There are several limitations to our study that readers should take into consideration. First, this study relies on retrospective data for breastfeeding,

child-care, and employment behaviors, which may be subject to recall error. However, we would expect this type of error to occur in both the high- and low-cash gift groups, which would affect the precision of our estimates but not bias our results. Second, if reported intentions do not reflect underlying parenting preferences, then it is possible that the income boost experienced by those in the high-cash gift group enabled mothers to meet their preferences; however, since we do not measure preferences, we cannot test this possibility. Understanding how parenting preferences, intentions, and behaviors regarding child investments (e.g., child care) evolve over time and how their evolution is influenced by social, cultural, and economic factors is important for designing policy. However, we found limited research in this area—with a few notable exceptions, mostly in the domain of child care (e.g., Riley and Glass 2002; Rose and Elicker 2010).

In addition, results related to mothers meeting their intentions should be interpreted as the impact of an unconditional and unanticipated cash transfer. Mothers may have responded differently had they been able to plan for the additional income prior to the birth of their infant. For example, women in the high-cash gift group may have been able to arrange child care and therefore delay returning to work if they had known about the cash gift beforehand. Future research should investigate whether anticipated cash transfers that begin at or before the time of birth affect care practices and maternal employment in the first year of life.

Taken together, the findings from this study suggest that a reliable, regular, but modest unconditional cash transfer is unlikely to shift how low-income mothers organize work, breastfeeding, and child care in their infant's first year of life in any substantial way. Although it is possible that more generous income supports could influence care practices, our null results point to the complexity of caring for young children and underscore how parenting decisions are also shaped by noneconomic factors such as cultural norms, historical and current structural racism, and personal beliefs. The availability of supports—including breastfeeding assistance (e.g., lactation consultants, home-visiting programs), high-quality child care, and paid leave—likely also plays a key role, especially for low-income communities of color. Future work should explore whether access to these factors, in combination with cash transfers, meaningfully affect parenting practices and allow parents to better meet their parenting goals. In addition, there are many other aspects of family life that may be positively affected by these cash transfers that are not considered in this article.

NOTE

Laura Stilwell is an MD/PhD candidate studying public policy at Duke University. She successfully defended her PhD dissertation in spring 2024 and is now in her final year of medical school.

Maritza Morales-Gracia is an assistant professor in the Department of Child and Family Studies at California State University, Los Angeles.

Katherine Magnuson is the director of the Institute for Research on Poverty and a Vilas Distinguished Professor at the Sandra Rosenbaum School of Social Work, both at University of Wisconsin–Madison.

Lisa A. Gennetian is the Pritzker Professor of Early Learning Policy Studies at the Sanford School of Public Policy, Duke University.

Maria Sauval is a PhD candidate at the School of Education, University of California, Irvine.

Nathan A. Fox is a Distinguished University Professor in the Department of Human Development and Quantitative Methodology, Program in Neuroscience and Cognitive Science, University of Maryland, College Park.

Sarah Halpern-Meehin is the Vaughan Bascom Professor of Women, Family, and Community in the Department of Human Development and the La Follete School of Public Affairs, both at the University of Wisconsin–Madison.

Hirokazu Yoshikawa is the Courtney Sale Ross Professor of Globalization and Education at New York University's Steinhardt School of Culture, Education, and Human Development. He is also a university professor and codirector of the Global TIES for Children Center, both at NYU.

Kimberly G. Noble is a professor of neuroscience and education at Teachers College and director of the Neurocognition, Early Experience and Development Lab, both at Columbia University.

Research reported in this publication was supported by the Eunice Kennedy Shriver National Institute of Child Health and Human Development of the National Institutes of Health under Award Numbers R01HD087384 and 2R01HD087384. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. This research was additionally supported by the US Department of Health and Human Services, Administration for Children and Families, Office of Planning, Research and Evaluation; the National Institute of Mental Health; the Office of Behavioral and Social Sciences Research at the Office of the Director, National Institutes of Health; the Andrew and Julie Klingenstein Family Fund; the Annie E. Casey Foundation; Arnold Ventures; Arrow Impact; the Blue Cross and Blue Shield of Louisiana Foundation; the Bezos Family Foundation; the Bill and Melinda Gates Foundation; Bill Hammack and Janice Parmelee; the Brady Education Fund; the Chan Zuckerberg Initiative (Silicon Valley Community Foundation); the Charles and Lynn Schusterman Family Philanthropies; the Child Welfare Fund; the Esther A. and Joseph Klingenstein Fund; the Ford Foundation; the Greater New Orleans Foundation; the

Heising-Simons Foundation; the Holland Foundation; the Jacobs Foundation; the JPB Foundation; J-PAL North America; the Lozier Foundation; the New York City Mayor's Office for Economic Opportunity; the Perigee Fund; the Robert Wood Johnson Foundation; the Robin Hood Foundation; the Sherwood Foundation; the Valhalla Foundation; the Weitz Family Foundation; the W. K. Kellogg Foundation; and three anonymous donors. We thank the University of Michigan Survey Research Center, which acted as our partner in recruitment, data collection, and participant location and retention.

REFERENCES

- Aizer, Anna, Hilary Hoynes, and Adriana Lleras-Muney. 2022. "Children and the US Social Safety Net: Balancing Disincentives for Adults and Benefits for Children." *Journal of Economic Perspectives* 36 (2): 149–74.
- Ajzen, Icek. 1991. "The Theory of Planned Behavior." *Organizational Behavior and Human Decision Processes* 50 (2): 179–211.
- Anderson, Julie. 2016. "Quick Figures: Breadwinner Mothers by Race/Ethnicity and State." Institute for Women's Policy Research. Published September. <https://iwp.org/wp-content/uploads/2020/08/Q054.pdf>.
- Baby-Friendly USA. 2023. "About Us." <https://www.babyfriendlyusa.org/about/#:~:text=The%20BFHI%20assists%20hospitals%20in,hospitals%20that%20have%20done%20so.>
- Baker, Michael, and Kevin Milligan. 2008. "Maternal Employment, Breastfeeding, and Health: Evidence from Maternity Leave Mandates." *Journal of Health Economics* 27 (4): 871–87.
- Berger, Lawrence M., Jennifer Hill, and Jane Waldfogel. 2005. "Maternity Leave, Early Maternal Employment and Child Health and Development in the US." *Economic Journal* 115 (501): F29–F47.
- Bitler, Marianne, Hilary Hoynes, and Elira Kuka. 2017. "Child Poverty, the Great Recession, and the Social Safety Net in the United States." *Journal of Policy Analysis and Management* 36 (2): 358–89.
- Blau, David, and Janet Currie. 2006. "Pre-school, Day Care, and After-School Care: Who's Minding the Kids?" *Handbook of the Economics of Education* 2:1163–278.
- Borra, Cristina, Maria Iacovou, and Almudena Sevilla. 2015. "New Evidence on Breastfeeding and Postpartum Depression: The Importance of Understanding Women's Intentions." *Maternal and Child Health Journal* 19:897–907.
- Bronfenbrenner, Urie, and Pamela A. Morris. 2007. "The Bioecological Model of Human Development." In *Handbook of Child Psychology: Theoretical Models of Human Development*. 6th ed., edited by William Damon and Richard M. Lerner, 793–828. Hoboken, NJ: Wiley. <https://onlinelibrary.wiley.com/doi/10.1002/9780470147658.chpsy0114>.
- Brooks-Gunn, Jeanne, Wen-Jui Han, and Jane Waldfogel. 2002. "Maternal Employment and Child Cognitive Outcomes in the First Three Years of Life: The NICHD Study of Early Child Care." *Child Development* 73 (4): 1052–72.
- CDC (Centers for Disease Control and Prevention). 2023. "Breastfeeding among U.S. Children Born 2013–2020, CDC National Immunization Survey—Child." https://www.cdc.gov/breastfeeding/data/nis_data/results.html.

- Chai, Yan, Arijit Nandi, and Jody Heymann. 2018. "Does Extending the Duration of Legislated Paid Maternity Leave Improve Breastfeeding Practices? Evidence from 38 Low-Income and Middle-Income Countries." *BMJ Global Health* 3 (5): e001032.
- Chatterji, Pinka, and Kevin D. Frick. 2005. "Does Returning to Work after Childbirth Affect Breastfeeding Practices?" *Review of Economics of the Household* 3 (3): 315–35.
- Chaudry, Ajay. 2004. *Putting Children First: How Low-Wage Working Mothers Manage Child Care*. New York: Russell Sage.
- Child Care Aware of America. 2022. "Demanding Change: Repairing Our Child Care System." [https://info.childcareaware.org/hubfs/2022-03-FallReport-FINAL%20\(1\).pdf?utm_campaign=Budget%20Reconciliation%20Fall%202021&utm_source=website&utm_content=22_demandingchange_pdf_update332022](https://info.childcareaware.org/hubfs/2022-03-FallReport-FINAL%20(1).pdf?utm_campaign=Budget%20Reconciliation%20Fall%202021&utm_source=website&utm_content=22_demandingchange_pdf_update332022).
- Constantino, Sara M., and Emmett Zeifman. 2021. "Cash Transfers, UBI and a Changing Political Landscape." *Ethical Perspectives* 28 (1): 63–75.
- Crosby, Danielle, and Julia Mendez. 2016. "Hispanic Children's Participation in Early Care and Education: Amount and Timing of Hours by Household Nativity Status, Race/Ethnicity, and Child Age." National Research Center on Hispanic Children and Families. Published November. <https://www.hispanicresearchcenter.org/wp-content/uploads/2019/08/NSECE-Series-HoursAmtTiming-V21.pdf>.
- Davidson, Eirwyn L., and Richard L. Ollerton. 2020. "Partner Behaviours Improving Breastfeeding Outcomes: An Integrative Review." *Women and Birth* 33 (1): e15–e23.
- DeVane-Johnson, Stephanie, Cheryl Woods Giscombe, Ronald Williams II, Cathie Fogel, and Suzanne Thoyre. 2018. "A Qualitative Study of Social, Cultural, and Historical Influences on African American Women's Infant-Feeding Practices." *Journal of Perinatal Education* 27 (2): 71–85.
- Dinour, Lauren M., and Yeon K. Bai. 2016. "Breastfeeding: The Illusion of Choice." *Women's Health Issues* 26 (5): 479–82.
- Enchautegui, María E., Martha Johnson, and Julia Gelatt. 2015. *Who Minds the Kids When Mom Works a Nonstandard Schedule?* Washington, DC: Urban Institute.
- Fitzsimons, Emla, and Marcos Vera-Hernández. 2022. "Breastfeeding and Child Development." *American Economic Journal: Applied Economics* 14 (3): 329–66.
- Gennetian, Lisa A., A. Rupa Datta, Robert Goerge, Wladimir Zanon, Richard Brandon, Ann Witte, and Parvati Krishnamurty. 2019. "How Much of Children's Time in Nonparental Care Coincides with Their Parents' Time at Work?" *Socius* 5:237802311989484.
- Gennetian, Lisa A., Greg Duncan, Nathan A. Fox, Katherine Magnuson, Sarah Halpern-Meekin, Kimberly G. Noble, and Hirokazu Yoshikawa. 2022. "Unconditional Cash and Family Investments in Infants: Evidence from a Large-Scale Cash Transfer Experiment in the U.S." Working Paper no. w30379. National Bureau of Economic Research, Cambridge, MA.
- Gennetian, Lisa A., and Anna Gassman-Pines. 2022. "Parents as Earners: What Parental Work Means for Parenting and the Role of Public Policy." In *The Cambridge Handbook of Parenting*, edited by Amanda S. Morris and Julia M. Smith, 536–59. New York: Cambridge University Press.
- Gennetian, Lisa A., Sarah Halpern-Meekin, Lauren Meyer, Nathan Fox, Katherine Magnuson, Kimberly G. Noble, and Hirokazu Yoshikawa. 2023. "Cash to U.S. Families at Scale: Behavioral

- Insights on Implementation from the Baby's First Years Study." In *Using Cash Transfers to Build an Inclusive Society: A Behaviorally Informed Approach*, edited by Jiaying Zhao, Saugato Datta, and Dilip Soman, 193–216. Toronto: University of Toronto Press.
- Gibson-Davis, Christina M., and Jeanne Brooks-Gunn. 2006. "Couples' Immigration Status and Ethnicity as Determinants of Breastfeeding." *American Journal of Public Health* 96 (4): 641–46.
- Gilbert, Melissa R. 1998. "'Race,' Space, and Power: The Survival Strategies of Working Poor Women." *Annals of the Association of American Geographers* 88 (4): 595–621.
- Gill, Sara L. 2009. "Breastfeeding by Hispanic Women." *Journal of Obstetric, Gynecologic and Neonatal Nursing* 38 (2): 244–52.
- Goldin, Claudia. 1977. "Female Labor Force Participation: The Origin of Black and White Differences, 1870 and 1880." *Journal of Economic History* 37 (1): 87–108.
- Goldin, Claudia, Sari Pekkala Kerr, and Claudia Olivetti. 2022. "When the Kids Grow Up: Women's Employment and Earnings across the Family Cycle." Working Paper no. w30323. National Bureau of Economic Research, Cambridge, MA.
- Green, Victoria L., Nekisha L. Killings, and Camille A. Clare. 2021. "The Historical, Psycho-social, and Cultural Context of Breastfeeding in the African American Community." *Breastfeeding Medicine* 16 (2): 116–20.
- Guttman, Nurit, and Deena R. Zimmerman. 2000. "Low-Income Mothers' Views on Breastfeeding." *Social Science and Medicine* 50 (10): 1457–73.
- Halpern-Meehin, Sarah, Lisa A. Gennetian, Jill Hoiting, Laura Stilwell, and Lauren Meyer. 2024. "Monthly Unconditional Income Supplements Starting at Birth: Experiences among Mothers of Young children in Poverty in the U.S." *Journal of Policy Analysis and Management*. <https://doi.org/10.1002/pam.22571>.
- Henly, Julia R., and Susan J. Lambert. 2014. "Unpredictable Work Timing in Retail Jobs: Implications for Employee Work-Life Conflict." *ILR Review* 67 (3): 986–1016.
- Herbst, Chris M. 2017. "Are Parental Welfare Work Requirements Good for Disadvantaged Children? Evidence from Age-of-Youngest-Child Exemptions." *Journal of Policy Analysis and Management* 36 (2): 327–57.
- Hofferth, Sandra L., Kimberlee A. Shauman, Robin R. Henke, and Jerry West. 1998. *Characteristics of Children's Early Care and Education Programs: Data from the 1995 National Household Education Survey*. Washington, DC: National Center for Education Statistics.
- Hoynes, Hilary W., and Diane Whitmore Schanzenbach. 2018. "Safety Net Investments in Children." Working Paper no. w24594. National Bureau of Economic Research, Cambridge, MA.
- Johnson, Anna D. 2017. "Child Care and Child Development in the United States: Where Have We Come from, What Do We Know Now, and Where Are We Going?" In *The Wiley Handbook of Early Childhood Development Programs, Practices, and Policies*, edited by Elizabeth Votruba-Drzal and Eric Dearing, 261–85. Hoboken, NJ: Wiley.
- Johnson, Katherine M. 2019. "Making Milk and Money: Contemporary Mothers' Orientations to Breastfeeding and Work." *Sociological Forum* 34 (4): 883–903.
- Jones, Katherine M., Michael L. Power, John T. Queenan, and Jay Schulkin. 2015. "Racial and Ethnic Disparities in Breastfeeding." *Breastfeeding Medicine* 10 (4): 186–96.

- Katz, Michael B. 2013. *The Undeserving Poor: America's Enduring Confrontation with Poverty*. 2nd ed. New York: Oxford University Press.
- Kim, Soojung, and Andrew D. Williams. 2023. "Roles of Income and Acculturation in the Hispanic Paradox: Breastfeeding among Hispanic Women." *Maternal and Child Health Journal* 27 (6): 1070–80.
- Kimmel, Jean, and Rachel Connelly. 2007. "Mothers' Time Choices Caregiving, Leisure, Home Production, and Paid Work." *Journal of Human Resources* 42 (3): 643–81.
- Knop, Brian. 2019. "Among Recent Moms, More Educated Most Likely to Work." US Census Bureau. Published August 19. <https://www.census.gov/library/stories/2019/08/are-women-really-opting-out-of-work-after-they-have-babies.html>.
- Laughlin, Lynda Lvonne. 2011. "Maternity Leave and Employment Patterns of First-Time Mothers: 1961–2008." US Census Bureau. Published October. <https://www2.census.gov/library/publications/2011/demo/p70-128.pdf>.
- Liang, Xiaoyan, Bruce Fuller, and Judith D. Singer. 2000. "Ethnic Differences in Child Care Selection: The Influence of Family Structure, Parental Practices, and Home Language." *Early Childhood Research Quarterly* 15 (3): 357–84.
- Lind, Jennifer N., Cria G. Perrine, Ruowei Li, and Kelley S. Scanlon, and Laurence M. Grummer-Strawn. 2014. "Racial Disparities in Access to Maternity Care Practices That Support Breastfeeding—United States, 2011." *Morbidity and Mortality Weekly Report* 63 (33): 725–28.
- Livingston, Gretchen. 2018. "About One-Third of U.S. Children Are Living with an Unmarried Parent." Pew Research Center. Published April 27. <https://www.pewresearch.org/short-reads/2018/04/27/about-one-third-of-u-s-children-are-living-with-an-unmarried-parent/>.
- Lowe, Edward D., and Thomas S. Weisner. 2004. "'You Have to Push It—Who's Gonna Raise Your Kids?' Situating Child Care and Child Care Subsidy Use in the Daily Routines of Lower Income Families." *Children and Youth Services Review* 26 (2): 143–71.
- Lubbers, Miranda J., Mario Luis Small, and Hugo Valenzuela García. 2020. "Do Networks Help People to Manage Poverty? Perspectives from the Field." *Annals of the American Academy of Political and Social Science* 689 (1): 7–25.
- Magnuson, Katherine, and Hilary Shager. 2010. "Early Education: Progress and Promise for Children from Low-Income Families." *Children and Youth Services Review* 32 (9): 1186–98.
- Marinescu, Ioana. 2018. "No Strings Attached: The Behavioral Effects of US Unconditional Cash Transfer Programs." Working Paper no. w24337. National Bureau of Economic Research, Cambridge, MA.
- Meek, Joan Y., and Lawrence Noble. 2022. "Technical Report: Breastfeeding and the Use of Human Milk." *Pediatrics* 150 (1): e2022057989.
- Meyers, Marcia K., and Lucy P. Jordan. 2006. "Choice and Accommodation in Parental Child Care Decisions." *Community Development* 37 (2): 53–70.
- Mirkovic, Kelsey R., Cria G. Perrine, and Kelley S. Scanlon. 2016. "Paid Maternity Leave and Breastfeeding Outcomes." *Birth* 43 (3): 233–39.
- National Survey of Early Care and Education Project Team. 2021. "2019 NSECE Snapshot: Parent Work Schedules in Households with Young Children." OPRE Report no. 2021-187. Office of Planning, Research and Evaluation, US Department of Health and Human Services, Washington DC.

- NCSL (National Conference of State Legislatures). 2021. "Breastfeeding State Laws." <https://www.ncsl.org/health/breastfeeding-state-laws>.
- New York State Department of Health. 2017. "For Breastfeeding Mothers Going Back to Work." https://www.health.ny.gov/prevention/nutrition/wic/breastfeeding/back_to_work_moms.htm#:~:text=Under%20NYS%20Labor%20Law%20Section,three%20years%20after%20giving%20birth.
- Noble, Kimberly G., Katherine Magnuson, Lisa A. Gennetian, Greg J. Duncan, Hirokazu Yoshikawa, Nathan A. Fox, and Sarah Halpern-Meekin. 2021. "Baby's First Years: Design of a Randomized Controlled Trial of Poverty Reduction in the United States." *Pediatrics* 148 (4): e2020049702.
- NYS (New York State). 2023. "Paid Family Leave Fact Sheet." https://www.ny.gov/sites/ny.gov/files/atoms/files/PaidFamilyLeave_EmployeeFactSheet.pdf.
- Pac, Jessica E., Ann P. Bartel, Christopher J. Ruhm, and Jane Waldfogel. 2019. "Paid Family Leave and Breastfeeding: Evidence from California." Working Paper no. w25784. National Bureau of Economic Research, Cambridge, MA.
- Pac, Jessica, and Lawrence M. Berger. 2024. "Quasi-Experimental Evidence on the Employment Effects of the 2021 Fully Refundable Monthly Child Tax Credit." *Journal of Policy Analysis and Management* 43 (1): 192–213.
- Pilkaukas, Natasha, Katherine Micheltore, Nicole Kovski, and H. Luke Shaefer. 2022. "The Effects of Income on the Economic Wellbeing of Families with Low Incomes: Evidence from the 2021 Expanded Child Tax Credit." Working Paper no. w30533. National Bureau of Economic Research, Cambridge, MA.
- Prenatal-to-3 Policy Impact Center. 2022. "Prenatal-to-3 Policy Clearinghouse Evidence Review: Child Care Subsidies (ER 07C.1022)." <http://pn3policy.org/policy-clearinghouse/child-care-subsidies>.
- Riley, Lisa A., and Jennifer L. Glass. 2002. "You Can't Always Get What You Want: Infant Care Preferences and Use among Employed Mothers." *Journal of Marriage and Family* 64 (1): 2–15.
- Rose, Katherine Kensinger, and James Elicker. 2010. "Maternal Child Care Preferences for Infants, Toddlers, and Preschoolers: The Disconnect between Policy and Preference in the USA." *Community, Work and Family* 13 (2): 205–29.
- Sauval, Maria, Greg Duncan, Lisa A. Gennetian, Katherine Magnuson, Nathan Fox, Kimberly Noble, and Hirokazu Yoshikawa. 2022. "Unconditional Cash Transfers and Maternal Employment: Evidence from the Baby's First Years Study." SSRN. <https://doi.org/10.2139/ssrn.4297310>.
- Shuey, Elizabeth A., and Tama Leventhal. 2020. "Enriched Early Childhood Experiences: Latina Mothers' Perceptions and Use of Center-Based Child Care." *Early Childhood Research Quarterly* 52:49–62.
- Smith, Sandra S. 2016. "Job-Finding among the Poor: Do Social Ties Matter?" In *The Oxford Handbook of the Social Science of Poverty*, edited by David Brady and Linda M. Burton, 438–61. New York: Cambridge University Press.
- Stanford Basic Income Lab. 2023. "Research: Experiments Map." <https://basicincome.stanford.edu/experiments-map/>.
- Stein, Gabriela L., Alexandra M. Cupito, Julia L. Mendez, Juan Prandoni, Nadia Huq, and Diana Westerberg. 2014. "Familism through a Developmental Lens." *Journal of Latina/o Psychology* 2 (4): 224–50.

- Stilwell, Laura, and Lisa Gennetian. 2022. "Economics of Infant Feeding in the U.S." Published October 7. <https://econofact.org/economics-of-infant-feeding-in-the-u-s#:~:text=Assuming%20an%20infant%20breastfeeds%20for,in%20the%20first%20month%20alone>.
- Swenson, Kendall, and Kimberly Burgess Simms. 2020. "Early Care and Education Arrangements of Children under Age Five." US Department of Health and Human Services. Published July. https://aspe.hhs.gov/sites/default/files/migrated_legacy_files/195451/Preschoolers_Type_of_Care.pdf.
- US Department of Labor. 2023a. "Fact Sheet #73: FLSA Protections for Employees to Pump Breast Milk at Work." Revised January. <https://www.dol.gov/agencies/whd/fact-sheets/73-flsa-break-time-nursing-mothers>.
- . 2023b. "Section 7(r) of the Fair Labor Standards Act—Break Time for Nursing Mothers Provision." [https://www.dol.gov/agencies/whd/nursing-mothers/law#:~:text=Nursing%20Mothers%20Provision-,Section%207\(r\)%20of%20the%20Fair%20Labor%20Standards%20Act%20%E2%80%93,the%20birth%20of%20her%20child](https://www.dol.gov/agencies/whd/nursing-mothers/law#:~:text=Nursing%20Mothers%20Provision-,Section%207(r)%20of%20the%20Fair%20Labor%20Standards%20Act%20%E2%80%93,the%20birth%20of%20her%20child).
- Weisner, Thomas S. 2002. "Ecocultural Understanding of Children's Developmental Pathways." *Human Development* 45 (4): 275–81.
- Westfall, Peter H., and S. Stanley Young. 1993. *Resampling-Based Multiple Testing: Examples and Methods for p-Value Adjustment*. New York: Wiley.
- Wolf, Jacqueline H. 2003. "Low Breastfeeding Rates and Public Health in the United States." *American Journal of Public Health* 93 (12): 2000–2010.
- Yearby, Ruqaiijah. 2020. "Employment Discrimination, Breastfeeding, and Health Justice." *California Western Law Review* 57 (2): 279–89.